Forensic Dump v2.0.1

Complete Step-by-Step User & Technical Reference Manual

Digital Forensics Investigation Suite

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1 Introduction

Forensic Dump v2.0.1 is a high-performance C-based software utility utilizing the GTK+3 graphical toolkit. It is designed to assist forensic investigators, cybersecurity analysts, and system administrators in extracting, carving, and parsing storage media structures. By employing a multi-threaded backend alongside a clean, modern user interface, the tool ensures high responsiveness and safety when analyzing large disk volumes.

The primary capabilities of Forensic Dump include:

- **Bit-Stream Partition Acquisition:** Byte-by-byte duplication of live storage media blocks into flat raw image files.
- **Magic-Byte File Carving:** Reconstructing file entities (JPEG, PNG, PDF, ZIP) without relying on partition table metadata or filesystem allocation tables.
- **Credential & Signature Scanning:** Locating critical textual patterns such as PEM-formatted cryptographic private keys, email addresses, high-entropy password strings, and custom search keywords.

2 System Prerequisites & Installation

2.1 Prerequisites

To compile and execute the application, you must install the GCC compiler, GNU Make, GTK+3 development libraries, GLib, and Pkg-config. On Debian, Ubuntu, or derivative distributions, run:

```
sudo apt-get update
sudo apt-get install build-essential libgtk-3-dev pkg-config libglib2
    .0-dev
```

2.2 Compilation

The source code contains a standard Makefile configured to compile ‘src/main.c‘ and ‘src/forensics.c‘ into a single standalone binary named ‘forensic-tool‘. To build, run:

```
make
```

This will create a subdirectory ‘obj/‘ for intermediate objects and the final executable ‘forensic-tool‘ in the project root folder.

2.3 Global Installation

To install the application globally, deploying desktop configuration files, application icons, and linking with system launcher services:

```
sudo make install
```

This command deploys:

- The binary to `/usr/local/bin/forensic-tool`
- The desktop launcher file to `/usr/share/applications/forensic-tool.desktop`
- Application PNG icons to `/usr/share/icons/hicolor/scalable/apps/` and `/usr/share/pixmaps/`

2.4 Uninstallation

To remove the application and refresh the desktop databases, use:

```
sudo make uninstall
```

3 Step-by-Step Graphical User Interface Guide

The Forensic Dump interface is divided into three distinct tabs corresponding to the utility's core modules. The application includes a custom dark theme (dark slate background, styled buttons, and distinct text fields) to reduce eye strain during prolonged forensic reviews. Below is a detailed, button-by-button guide to using each tab.

3.1 Acquire (Dump Partition) Tab

This tab is used to copy a live physical partition or raw block device (e.g., an external USB drive) into a raw flat image file.

3.1.1 Workflow and UI Controls

1. **Select Source Device:** Locate the field labeled **Source Device / File:**.
   - Either type the block device path directly (e.g., `/dev/sdb1`) into the entry box,
   - Or click the **Browse / Dev** button on the right to open a system file chooser and browse directly to a device block descriptor.
2. **Specify Output File:** Locate the field labeled **Destination Dump (.img):**.
   - Click the **Select Destination** button on the right to open a save file dialog.
   - Choose the target directory and name the dump file (e.g., `usb_dump.img`).
3. **Execute Acquisition:** Click the blue **Start Acquisition** button at the bottom of the page.
   - The application will spawn a background duplication thread.
   - The status indicator below the progress bar will update from "Ready." to a active percentage and transfer tally (e.g., `Acquiring:  512.0 MB / 2048.0 MB (25.0%)`).
4. **Cancellation Option:** While running, the **Start Acquisition** button will be disabled, and a red **Cancel** button will appear. If you need to stop the transfer, click the **Cancel** button immediately to safely flush buffers and abort the job.

3.1.2 Step-by-Step Acquisition Example

To acquire a partition named `/dev/sdc1` and save it to `/home/user/evidence.img`:

1. Launch the tool with administrative privileges: `sudo forensic-tool` (required to read `/dev/sdc1`).
2. Under the **Acquire (Dump)** tab, click the first input text entry and type `/dev/sdc1`.
3. Click **Select Destination**, navigate to `/home/user/`, type `evidence.img` in the name field, and click **Save**.
4. Click **Start Acquisition**.
5. Monitor the progress bar. Once completed, the status label will display: `Dump acquired successfully`.

3.2 File Carving Tab

This tab scans raw dump files (such as those generated by the Acquire module) for magic signatures to extract deleted or orphaned files.

3.2.1 Workflow and UI Controls

1. **Select Dump File:** Locate the field labeled **Dump Image File:**.
   - Click the **Select Dump** button on the right to open a file browser.
   - Navigate to your raw image file (e.g., `usb_dump.img`) and click **Open**.
2. **Choose Output Folder:** Locate the field labeled **Output Folder:**.
   - Click the **Select Output Folder** button on the right.
   - Select a destination directory where carved files will be saved (e.g., `/home/user/carved_output/`) and click **Select**.
3. **Configure File Signatures:** Use the checkboxes in the configuration row to toggle file formats on or off:
   - **Carve JPEG (.jpg)**: Toggle to extract JPEG images.
   - **Carve PNG (.png)**: Toggle to extract PNG images.
   - **Carve PDF (.pdf)**: Toggle to extract PDF files.
   - **Carve ZIP (.zip)**: Toggle to extract ZIP files (also includes Office XML files like DOCX/XLSX).
4. **Execute Carving:** Click the blue **Start File Carving** button at the bottom of the tab.
   - The background loop begins scanning.
   - Any reconstructed file is immediately appended to the results table showing the **Filename**, its **Offset (Bytes)** location in the image, and the computed **Size (Bytes)**.
5. **Cancellation Option:** Click the red **Cancel** button to halt carving at any time.

3.2.2 Step-by-Step Carving Example

To carve PNG and PDF assets from `/home/user/evidence.img`:

1. Select the **File Carving** tab.
2. Click **Select Dump**, select `/home/user/evidence.img`, and click **Open**.
3. Click **Select Output Folder**, select the directory `/home/user/carved/`, and click **Select**.
4. Uncheck **Carve JPEG (.jpg)** and **Carve ZIP (.zip)**. Leave **Carve PNG** and **Carve PDF** checked.
5. Click **Start File Carving**.
6. The table will populate with items such as `carved_409600.png`.
7. Wait for the status label to read: `Carving scan completed.` then check the output folder.

3.3 Credential / Key Search Tab

This tab parses a forensic image to search for plain-text patterns, credentials, or keys.

3.3.1 Workflow and UI Controls

1. **Select Dump File:** Locate the field labeled **Dump Image File:**.
   - Click the **Select Dump** button on the right, pick the target image, and click **Open**.
2. **Define Search Filters:** Check the specific patterns you wish to target:
   - **Private Keys (PEM)**: Searches for `-----BEGIN` headers.
   - **Email Addresses**: Checks for common address structures.
   - **Passwords / Credentials**: Scans printable strings matching high-entropy complexity rules.
3. **Add Custom Search Term (Optional):** Locate the field labeled **Custom Term Search:**.
   - Type any specific word or identifier (e.g., `confidential`, `API_KEY`, or a username) to execute a case-insensitive substring match.
4. **Execute Search:** Click the blue **Start Search** button at the bottom of the tab.
   - Found results populate the table showing the **Match Type**, a **Matched Text Preview** (up to 120 characters), and the decimal **Offset (Bytes)** of the match.

3.3.2 Step-by-Step Search Example

To look for private cryptographic files and custom words like "flag" inside an image:

1. Navigate to the **Credential / Key Search** tab.
2. Click **Select Dump** and select `evidence.img`.
3. In the **Custom Term Search** text box, type `flag`.
4. Check **Private Keys (PEM)**. Uncheck **Email Addresses** and **Passwords / Credentials** to focus the scan.
5. Click the blue **Start Search** button.
6. Any matched strings will appear in the table. When finished, the status will show `Forensic search completed.`

4 Functional Reference Guide

4.1 Partition Acquisition (Acquire)

The **Acquire** tool allows investigators to read storage block devices (e.g., `/dev/sda1`, `/dev/sdb2`) or standard files and copy them bit-by-bit to a target raw dump (`.img`).

- **Security Warning:** Reading block devices typically requires raw access permissions. Running the application under root privileges (`sudo forensic-tool`) or adding the user to appropriate storage groups (e.g. `disk`) is mandatory for block device operations.
- **Size Detection:** The application utilizes the Linux-specific `ioctl` parameter `BLKGETSIZE64` to identify block storage limits. If the source is a flat file instead of a block device, it falls back to standard `lseek` boundary calculations.
- **Chunk Buffer size:** Data replication operates inside 1MB (`1024 * 1024` bytes) chunk allocations to maximize disk sequential throughput.

4.2 File Carving

File carving extracts files from disk images when filesystems are corrupted or deleted. The program scans the raw dump stream for predefined headers (magic bytes) and search trailers (end markers). The supported magic bytes structures include:

Format	Start Magic Bytes (Hex)	End Magic Bytes (Hex)
JPEG	`FF D8 FF`	`FF D9`
PNG	`89 50 4E 47 0D 0A 1A 0A`	`49 45 4E 44 AE 42 60 82`
PDF	`25 50 44 46` (`%PDF`)	`25 25 45 4F 46` (`%%EOF`)
ZIP	`50 4B 03 04` (`PK..`)	`50 4B 05 06` (EOCD Header)

Table 1: File signatures used by Forensic Dump.

- **Boundary Safety Overlap:** Standard buffer boundaries can slice a file signature in half. To prevent missing signatures, the carver employs a 4KB (`4096` bytes) overlap buffer window across block boundaries.
- **ZIP File EOCD Comment Correction:** Due to dynamic user metadata comments appended to ZIP structures, the carving function parses the final bytes of the End of Central Directory (EOCD) packet to dynamically compute comment-related offsets.
- **Limit Boundaries:** A safety cap of 50MB is enforced for carved assets to prevent runaway disk writes during corrupt signature scans.

4.3 Credential & Signature Search

Scans target files for sensitive strings, printable structures, and specific terms.

1. **Custom Keyword Search:** Performs a case-insensitive search for user-configured character strings.
2. **Private Keys (PEM):** Matches the header boundary string `-----BEGIN`.
3. **Email Patterns:** Analyzes text buffers against regular address layouts, ensuring logical placement of `@` and `.` separators.

4. **Credential Heuristics:** Identifies potential high-entropy credentials. Strings between 8 and 64 characters are assessed for complexity (checking if they contain uppercase, lowercase, numerical digits, and special characters) or typical sub-string tokens like `password`, `secret`, or `api_key`.

5 Technical Architecture & Thread Safety

5.1 Multi-Threading Model

Since scanning a raw block device or image can block the CPU for prolonged periods, the user interface remains interactive by delegating core logic to worker threads using the POSIX Threads (`pthread`) library.

- Worker threads are spawned using `pthread_create` and detached with `pthread_detach` to allow resource reclamation upon termination.
- GUI widgets (such as text lists, text boxes, and progress indicators) are **not thread-safe** and must not be directly manipulated by worker threads.

5.2 GUI Interaction via GLib Idle Loop

To bridge the gap between background operations and the GTK GUI event loop, Forensic Dump formats telemetry data into structural envelopes (e.g., `ProgressUpdate`, `CarveRowUpdate`) and queues them in the GTK main loop using:

```
g_idle_add(update_progress_idle, up);
```

The GLib scheduler runs these callbacks asynchronously within the context of the primary thread, preventing races, visual freezes, and SEGFAULT signals.

5.3 Cancellation Mechanism

Operations can be stopped safely using a global atomic flag:

```
static volatile int g_cancel_flag = 0;
```

When the user clicks the "Cancel" button, the GUI sets `g_cancel_flag = 1`. The active worker loops inspect this variable at the end of each chunk cycle, cleanly closing open file descriptors, freeing buffer memory, and terminating execution loops.

6 CSS Graphic Theme

The interface uses a customized Dark Mode theme applied via GTK's `GtkCssProvider`. The CSS incorporates a modern palette:

- **Background:** Dark Grey (`#111827`)
- **Component containers:** Slate Blue (`#1f2937`)
- **Highlights and Progress Indicators:** Accent Blue (`#3b82f6`)
- **Typography:** Inter, Outfit, or standard sans-serif fallbacks.